

Relax and Think.

BSc I/Analysis

Chennai Mathematical Institute

26-11-2025

Total Marks: 50

Semestral Examination

Time: 9:30 - 11:30

Instructions: Justify your steps. Write Clearly.

Results from class work: Use freely, quote precisely.

Result from Home-Assignments, if used, must be proved,

even if it is explained in class and hints given – because

I do not know whether you worked it out or not.

1. Show that for each $x \in \mathbb{R}$ the following series converges and defines a continuous function on $\mathbb{R}$. [10]

$$\sin x + 2 \sin \left(\frac{x}{3} \right) + 2^2 \sin \left(\frac{x}{3^2} \right) + 2^3 \sin \left(\frac{x}{3^3} \right) + 2^4 \sin \left(\frac{x}{3^4} \right) + \dots$$

2. Show for integers $m, n \geq 1$ [10]

$$\int_0^1 x^m (1-x)^n = \frac{m! n!}{(m+n+1)!}$$

3. Find the area of the bounded region enclosed between x -axis and the curve $y = 4 - x^2$. [10]

4. Test for convergence of the integral [10]

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx$$

5. Find the equation of tangent to the following curve at $x = 2$. [5]

$$y = \frac{8}{4 + x^2}$$

6. evaluate: [5]

$$\lim_{x \rightarrow 1} \left[\frac{x}{x-1} - \frac{1}{\log x} \right]$$

GOOD LUCK